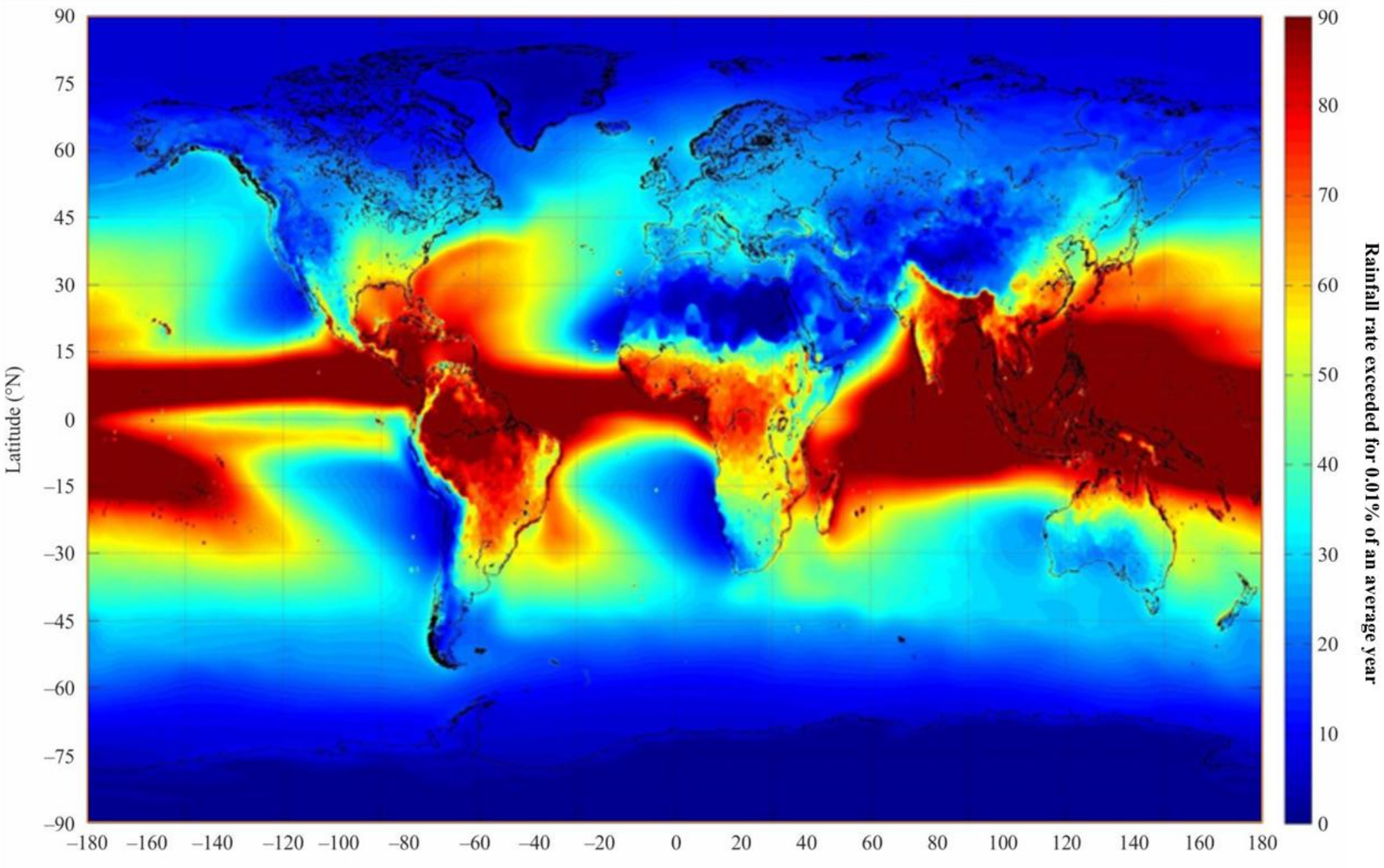
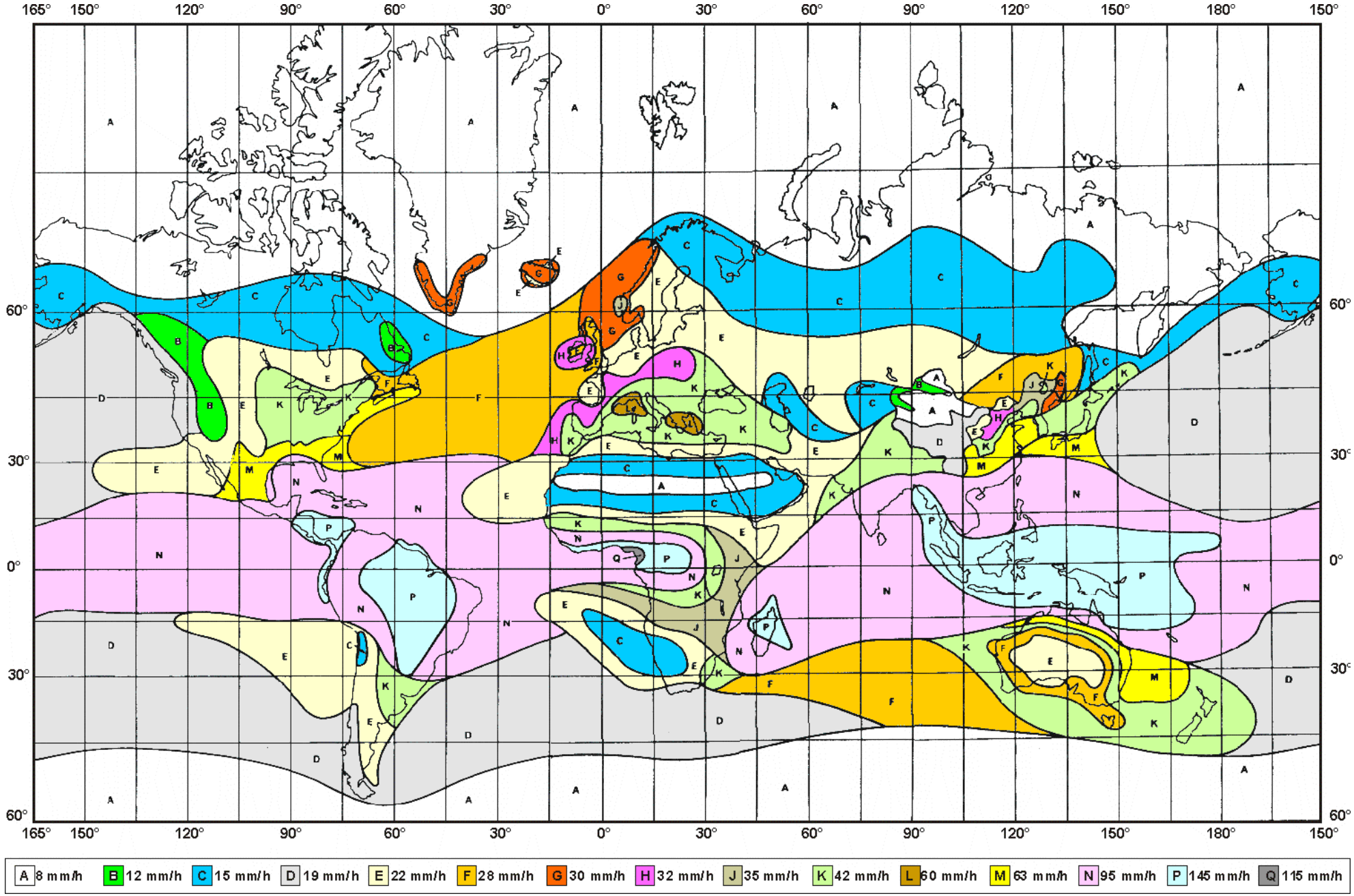
ITU Recommendation RREC-P.837-7-201706-I

https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.837-7-201706-S!!PDF-E.pdf




165° 150° 120° 90° 60° 30° 0° 30° 60° 90° 120° 150° 180° 150°
60°
30°
0°
30°
60°
A 8 mm/h
B 12 mm/h
C 15 mm/h
D 19 mm/h
E 22 mm/h
F 28 mm/h
G 30 mm/h
H 32 mm/h
J 35 mm/h
K 42 mm/h
L 60 mm/h
M 63 mm/h
N 95 mm/h
P 145 mm/h
Q 115 mm/h

Rain rate which is exceeded for a given percentage of time

A 8 mm/h | B 12 mm/h | C 15 mm/h | D 19 mm/h | E 22 mm/h | F 28 mm/h | G 30 mm/h | H 32 mm/h | J 35 mm/h | K 42 mm/h | L 60 mm/h | M 63 mm/h | N 95 mm/h | P 145 mm/h | Q 115 mm/h

Percentage of time (%)	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q
1.0	<0.1	0.5	0.7	2.1	0.6	1.7	3	2	8		2	4	5	12	14
0.3	0.8	2	2.8	4.5	2.4	4.5	7	4	13		7	11	15	34	49
0.1	2	3	5	8	6	8	12	10	20	12	15	22	35	65	72
0.03	5	6	9	13	12	15	20	18	28	23	33	40	65	105	96
0.01	8	12	15	19	22	28	30	32	35	42	60	63	95	145	115
0.003	14	21	26	29	41	54	45	55	45	70	105	95	140	200	142
0.001	22	32	42	42	70	78	65	83	55	100	150	120	180	250	170

E.g. South Europe

Then convert to specific attenuation in dB/km → Then compute dB over link exceeded with given probability

Other model: RECOMMENDATION ITU-R P.530-15
https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.530-16-201507-S!!PDF-E.pdf
→use data at 0.01% outage probability → compute specific attenuation and total attenuation
→ Then scale mathematically to required outage probability
Prop for MATLAB - File Exchange - MATLAB Central
https://it.mathworks.com/matlabcentral/fileexchange/116835-prop-for-matlab
rainfademargin(f_Hz,region_mmh,elevation_deg,polarization_deg_0H_90V,length_m,desiredAvailab_percent)
rainfademargin(18e9,42,0,90,15e3,99.9) %Returns 10.5dB

Step 1: Obtain the rain rate $R_{0.01}$ exceeded for 0.01% of the time (with an integration time of 1 min). If this information is not available from local sources of long-term measurements, an estimate can be obtained from the information given in Recommendation ITU-R P.837.

Step 2: Compute the specific attenuation, γ_R (dB/km) for the frequency, polarization and rain rate of interest using Recommendation ITU-R P.838.

Step 3: Compute the effective path length, d_{eff}, of the link by multiplying the actual path length d by a distance factor r. An estimate of this factor is given by:

$$r = \frac{1}{0.477\, d^{0.633} R_{0.01}^{0.073\alpha} f^{0.123} - 10.579(1-\exp(-0.024 d))} \quad (32)$$

where f (GHz) is the frequency and α is the exponent in the specific attenuation model from Step 2. Maximum recommended r is 2.5, so if the denominator of equation (32) is less than 0.4, use $r = 2.5$.

Step 4: An estimate of the path attenuation exceeded for 0.01% of the time is given by:

$$A_{0.01} = \gamma_R\, d_{eff} = \gamma_R\, d r \quad \text{dB} \quad (33)$$

Step 5: The attenuation exceeded for other percentages of time p in the range 0.001% to 1% may be deduced from the following power law:

$$\frac{A_p}{A_{0.01}} = C_1\, p^{-(C_2 + C_3 \log_{10} p)} \quad (34)$$

with:

$$C_1 = \left(0.07^{C_0}\right)\left[0.12^{(1-C_0)}\right] \quad (35a)$$

$$C_2 = 0.855 C_0 + 0.546(1 - C_0) \quad (35b)$$

$$C_3 = 0.139 C_0 + 0.043(1 - C_0) \quad (35c)$$

where:

$$C_0 = \begin{cases} 0.12 + 0.4\left[\log_{10}(f/10)^{0.8}\right] & f \geq 10\ GHz \\ 0.12 & f < 10\ GHz \end{cases} \quad (36)$$